KUMARESAN K

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PROFESSIONAL SUMMARY

Software Developer with hands-on experience building serverless and multi-tenant backend systems on AWS using Node.js/TypeScript, alongside a strong foundation in Python for data analysis, machine learning, and automation. Skilled in object-oriented design, debugging, version control, and agile collaboration. Combines production backend experience with a Computer Science (Data Science) background — including EDA, predictive modelling, and ML pipeline deployment (R^2 = 0.80) — to build clean, maintainable, well-tested software.

ACADEMIC QUALIFICATIONS

Bachelor of Engineering in Computer Science (Data Science) — Annamalai University | CGPA: 8.26 | Expected: May 2026

TECHNICAL SKILLS

Programming & Data: Python, TypeScript/Node.js, SQL, R, Pandas, NumPy

Backend & Cloud: NestJS, Next.js, AWS Lambda, API Gateway, Serverless Framework, Prisma, Drizzle ORM, PostgreSQL, Redis, Docker, AWS (Cognito, S3, ECS, CloudFormation, SAM, CodePipeline)

Data Visualization: Power BI (DAX, Power Query), Matplotlib, Seaborn

Machine Learning: Scikit-Learn, TensorFlow, Keras, Regression, Probabilistic Models

Statistics & Analytics: EDA, Customer Segmentation, Feature Engineering, Model Evaluation

Software Practices: Object-Oriented Design, Git/GitHub, Code Review, Unit Testing, Debugging, Agile

PROFESSIONAL EXPERIENCE

Associate Software Developer — Meyi Cloud Solutions Pvt. Ltd. (AWS Advanced Tier Partner)

Puducherry, India | May 2025 – Present

- **Current project – 7XT (multi-tenant radiology clinic RIS + PACS platform):** NestJS/TypeScript backend, Next.js UI, Prisma, PostgreSQL (Aurora), Redis, Cognito, ECS/ALB and S3.
- Built the typist workflow end to end: radiologist audio dictations stored in S3 (URL saved in DB) feed a typist worklist for transcription, followed by radiologist review, approve & sign or return-for-correction; signed reports are hashed for tamper detection. Also contributed to clinic implementation.
- Developed serverless backend components on AWS Lambda and API Gateway using Node.js/TypeScript, applying object-oriented and modular design principles for scalable, maintainable systems.
- Built and refactored a ticket management system with Drizzle ORM and PostgreSQL, moving business validation into the database layer using SQL CASE expressions and NULL-safe comparisons.
- Implemented application-level AES-256-GCM encryption with HMAC-SHA256 hashing to secure sensitive data in AWS RDS.
- Identified and resolved defects across authentication flows (AWS Cognito – registration, OTP, JWT), containerized local environments, and an activity-log module.
- Collaborated in an agile team using Git feature-branch workflows, multi-branch rebases and code reviews; used AWS SAM, CloudFormation and CodePipeline for CI/CD deployment scripting.

PROJECT EXPERIENCE

Customer Lifetime Value Prediction using Probabilistic Modelling | Python, Pandas, Lifetimes, Seaborn

- Developed BG/NBD + Gamma-Gamma CLV model on ~3,000 customers, identifying a VIP segment contributing ~90% of predicted revenue.
- Performed segmentation analysis to surface directly actionable insights for retention and paid-marketing strategies.

GitHub: github.com/kuman002/customer-lifetime-value-CLV

Bangalore House Price Prediction | Scikit-Learn, Flask, Python

- Built end-to-end ML pipeline: EDA → feature engineering → Linear Regression (R^2 = 0.80) → Flask REST API with real-time inference.
- Implemented structured logging, input validation, and exception handling for production readiness.

GitHub: github.com/kuman002/Bangalore-House-Price-Prediction

Intelligent Study Assistant (NLP & Information Retrieval) | Python, Vector Search, NLP

- Designed embedding-based semantic search system to retrieve contextually relevant academic content with source attribution.
- Implemented hallucination control via strict grounding prompts, returning explicit out-of-scope rejection when queries fall outside documents.

GitHub: github.com/kuman002/academic-notes-ai

Data Engineering Challenge (Slooze) | Web Scraping, Pandas, EDA

- Built automated web-scraping and data processing pipeline with Pandas; performed EDA to extract actionable insights from collected datasets.

GitHub: github.com/kuman002/slooze-take-home-challenge

CERTIFICATIONS

- IBM — Machine Learning for Data Science Projects (Mar 2026, Credly Verified)
- IBM — Enterprise Data Science in Practice (Feb 2026, Credly Verified)
- IBM — Data Fundamentals (Sep 2025, Credly Verified)

LEADERSHIP & ACTIVITIES

- Python Workshop Instructor — Delivered hands-on Python & data analysis curriculum for 30+ peers.
- Event Coordinator (COMPSEM) — Organized end-to-end logistics for a technical debate event.
- Open-Source GitHub — 36 public repos with consistent feature-branch workflow, meaningful commit messages, and MIT licences.